Week 9 Solution

NPTEL PSOSM Course (Privacy and Security in Online Social Media)

Question 1

Which of the following is/are location-based services? (multiple correct)

(a) Foursquare ✓

(b) Yelp ✓

(c) Gowalla ✓

(d) Uber ✓

Solution

A location-based service (LBS) is any service whose core function depends on the user's geographic location, usually taken from the phone's GPS.

Service	Role of location
Foursquare	Users 'check in' at venues; the lecture's main case study of a location-based social network
Yelp	Shows reviews and ratings of restaurants and businesses near the user
Gowalla	Check-in based social network (similar to Foursquare, now shut down); its check-in dataset is widely used in research
Uber	Uses the live location of both rider and driver to match them, route the trip and calculate the fare

All four would not work without the user's location, so all are location-based services.

Answer: Foursquare, Yelp, Gowalla and Uber (all four).

Question 2

What is the difference between Pearson and Spearman rank correlation?

(a) Pearson correlation assesses linear relationships, while Spearman correlation evaluates monotonic relationships. ✓

(b) Spearman correlation is always between -1 and 1, while Pearson correlation can have values outside this range.

(c) Spearman correlation is calculated using a formula involving the sum of squared differences, whereas Pearson correlation is based on the product of z-scores.

(d) All of the above

Solution

Pearson correlation measures how well the data fit a straight line:

$$r = \Sigma (x_i - \bar{x})(y_i - \bar{y}) / \sqrt{[\Sigma (x_i - \bar{x})^2 \cdot \Sigma (y_i - \bar{y})^2]}$$

Spearman correlation is Pearson's r applied to the ranks of the data. It measures whether y always increases (or always decreases) as x increases, which is a monotonic relationship, not necessarily a straight line. With no tied ranks:

$$\rho = 1 - 6 \Sigma d_i^2 / [n (n^2 - 1)], \quad \text{where } d_i = \text{difference between the ranks of } x_i \text{ and } y_i$$

Example: x = 1, 2, 3, 4, 5 and y = x^3 = 1, 8, 27, 64, 125. The relationship is perfectly increasing but curved. Spearman ρ = 1.00 while Pearson r ≈ 0.94 (both computed in Python). Spearman captures the perfect monotonic trend; Pearson is penalised because the points do not lie on a line.

Checking the other options:

- **(b) False:** both coefficients always lie between -1 and +1. Pearson can never go outside this range.
- **(c):** this describes how each is computed (the shortcut Spearman formula uses squared rank differences; Pearson equals the average product of z-scores), but it is not the conceptual difference the question asks for.
- **(d) False:** since (b) is false, 'All of the above' cannot be correct.

Answer: Pearson correlation assesses linear relationships, while Spearman correlation evaluates monotonic relationships.

Question 3

You have two datasets and want to measure the strength and direction of the relationship between them. One set consists of the ranks or ordinal values of the observations, while the other set consists of continuous numerical data. Which correlation coefficient is most appropriate?

(a) Spearman's rank correlation coefficient ✓

(b) Pearson's correlation coefficient

(c) Either Spearman or Pearson correlation can be used interchangeably.

(d) None of the above because correlation coefficients are not applicable in this scenario.

Solution

Pearson's r needs both variables to be continuous (interval or ratio scale), because it uses the actual distances between values. Ordinal data (ranks such as 1st, 2nd, 3rd, or ratings like low/medium/high) have an order, but the gaps between levels are not meaningful numbers.

Spearman's method works on ranks only. The continuous variable is converted into ranks, then the two rank lists are correlated. This uses only the ordering, which both datasets have, so Spearman is the right choice.

Example: correlating students' class rank (ordinal) with hours studied (continuous). Rank the hours, then compute ρ.

Option (c) is wrong because Pearson is not valid for ordinal data, and (d) is wrong because a rank correlation applies perfectly well here.

Answer: Spearman's rank correlation coefficient.

Question 4

Which of the following are open text fields, whose validity is not enforced by the system and may carry noise/invalid locations? (multiple correct)

(a) Venue location ✓

(b) Tips

(c) User home city ✓

(d) Dones

Solution

The lecture's Foursquare study collected location information from two fields that users type in freely:

- **Venue location:** the address or city of a venue, entered by the user who created the venue.
- **User home city:** typed in by the user on their profile.

Foursquare does not check these against a map, so they can contain spelling mistakes, abbreviations, jokes or invalid places (for example 'my home', 'Earth' or a misspelled city name). Researchers therefore had to clean them before using them for analysis.

Tips (short reviews or recommendations) and dones (marking that a tip was followed) are user activities attached to a venue, not location fields, so (b) and (d) are not the answer.

Answer: Venue location and User home city.

Question 5

Which of the following is the correct inference from Figure 2?

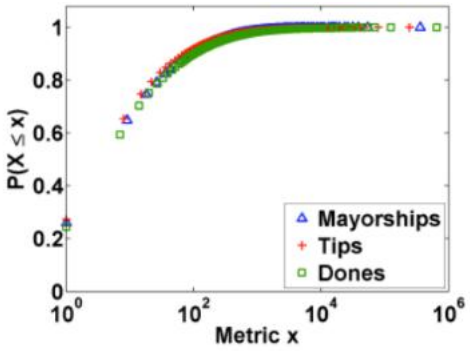


Figure 2. Cumulative Distribution of the Number of Mayorships, Tips and Dones per City.

(a) The distributions are very skewed, with a few cities having as many as 100 mayorships, tips or dones. ✓

(b) The distributions are evenly balanced, with cities having a nearly equal number of mayorships, tips, or dones.

(c) The distributions are normally distributed, with no cities having an exceptionally high number of mayorships, tips, or dones.

(d) None of the above

Solution

How to read a CDF: a point (x, y) means that a fraction y of all cities have a count of at most x. Note that the x-axis is logarithmic (1, 100, 10,000, 1,000,000).

Approximate readings for all three curves:

Value of x	P(X ≤ x)
1	about 0.25 (a quarter of cities have only 1)
about 10	about 0.65
about 100	about 0.95
10,000 and beyond	1.0 (only a handful of cities reach these values)

The curve rises very steeply at small x and then flattens into a long tail stretching across several orders of magnitude. Most cities have very few mayorships, tips or dones, while a few cities (large, popular ones) have 100 or more. This is a highly skewed, long-tailed distribution.

Option (b) would need the CDF to jump almost straight from 0 to 1 at one value, and (c) would need a symmetric S-shaped curve with no long tail; neither matches the figure.

Answer: The distributions are very skewed, with a few cities having as many as 100 mayorships, tips or dones.

Question 6

What is the key difference between a scatter plot and a bar chart?

(a) Scatter plots display categorical data, while bar charts are used for numerical data.

(b) Scatter plots show the relationship between two numerical variables, while bar charts display the frequency or distribution of categorical data. ✓

(c) Scatter plots are only used for displaying data with a single variable, while bar charts can visualize data with multiple variables.

(d) All of the above

Solution

	Scatter plot	Bar chart
X-axis	Numerical variable	Categories (cities, platforms, ...)
Y-axis	Numerical variable	Count, frequency or value per category
Each mark	A point = one observation (x, y)	A bar = one category
Used for	Relationship / correlation between two variables	Comparing amounts across categories
Example	Followers vs number of tweets per user	Number of users per social network

Option (a) reverses the roles, and (c) is wrong because a scatter plot needs two variables (one per axis).

Answer: Scatter plots show the relationship between two numerical variables, while bar charts display the frequency or distribution of categorical data.

Question 7

Which of the following is publicly available data on Foursquare? (multiple correct)

(a) Mayorships ✓

(b) Tips ✓

(c) Dones ✓

(d) Check-ins

Solution

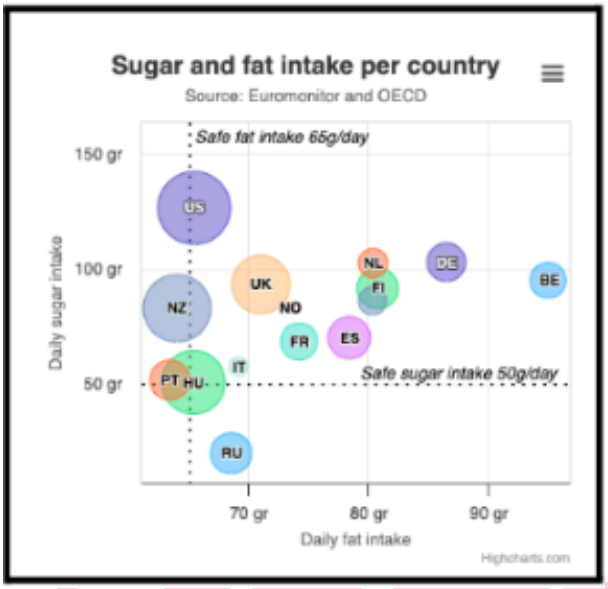
Data	Meaning	Visibility
Mayorships	The user with the most check-ins at a venue in the last 60 days becomes its 'mayor'	Public (shown on the venue page)
Tips	Short public reviews or recommendations about a venue	Public
Dones	Marks that a user has done what a tip suggested	Public
Check-ins	A user's record of being at a venue at a given time	Private by default (friends only)

Check-ins reveal exactly where a person is right now, so they are visible only to the user's friends and cannot be crawled publicly. This is why the lecture's study used mayorships, tips and dones, which anyone can see, as indirect evidence of where users had been.

Answer: Mayorships, Tips and Dones.

Question 8

Which plot is shown in the below figure?



(a) Scatter plot

(b) Bar chart

(c) Bubble plot ✓

(d) Histogram

Solution

The figure shows each country as a circle placed by daily fat intake (x-axis) and daily sugar intake (y-axis). Crucially, the circles have different sizes, which encodes a third variable (for example population or obesity rate).

A bubble plot is a scatter plot extended to three variables: x position, y position and bubble size. Here the US and NZ bubbles are much larger than RU or IT, so size is clearly carrying information. A plain scatter plot would use equal-sized points, and bar charts and histograms use bars, not points.

Answer: Bubble plot.

Question 9

What is the primary characteristic of Location-Based Social Networks (LBSNs)?

(a) They allow users to share text-based posts with their friends and followers.

(b) They provide a platform for online gaming and virtual reality experiences.

(c) They enable users to connect with others and share their physical locations and activities. ✓

(d) They focus exclusively on professional networking and job searching.

Solution

An LBSN adds a location layer to a social network. Users check in at real places, share where they are and what they are doing there, leave tips, and find friends or venues nearby. Foursquare, Gowalla and Swarm are examples.

Option (a) describes any social network (such as Twitter), (b) describes gaming platforms, and (d) describes LinkedIn. Only (c) mentions physical location, the defining feature of an LBSN.

Answer: They enable users to connect with others and share their physical locations and activities.

Question 10

As discussed in the lecture, choose the correct inference(s) for the figures below (Figure 4 and Figure 5). Here mayorships (blue), Tips (red) and Dones (green). (multiple correct)

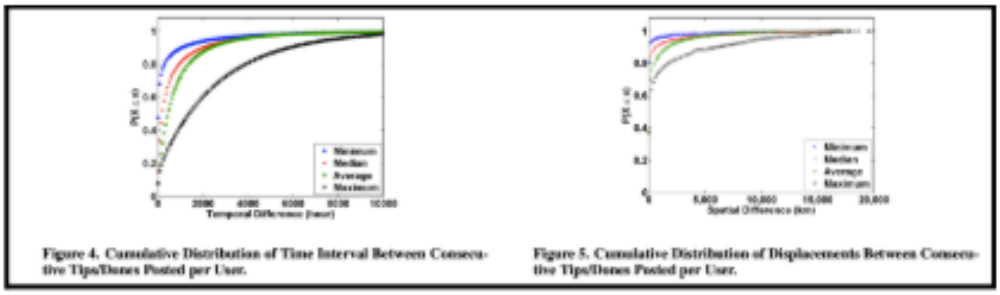


Figure 4. Cumulative Distribution of Time Interval Between Consecutive Tips/Dones Posted per User.

Figure 5. Cumulative Distribution of Displacements Between Consecutive Tips/Dones Posted per User.

(a) In Figure 4, the distribution of minimum inter-activity times is very skewed towards short periods of time, with almost 50% of the users posting consecutive tips/dones 1 hour apart. ✓

(b) In Figure 4, on average, median and maximum, users do tend to experience very long periods of time between consecutive tips and dones. For instance, around 50% of the users have an average inter-activity time of at least 450 hours, whereas around 80% of the users have a maximum inter-activity time above 167 hours (roughly a week). ✓

(c) In Figure 5, around 36% of the users have average and maximum displacements of 0 kilometer, indicating very short distances (within a few metres). ✓

(d) In Figure 5, about 10% of the users have a maximum displacement of at least 6,000 kilometers. ✓

Solution

Both figures come from the 2012 Foursquare privacy study given as the reference in the question (Precog group, IIIT Delhi, TP_lbsn_2012). For each user, the consecutive tips/dones they posted are compared, and four statistics are plotted as CDFs: minimum, median, average and maximum.

Figure 4: time between consecutive activities (hours).

- **(a) True:** the minimum curve shoots up almost immediately near x = 0, reaching about 0.5. So about half the users have, at least once, posted two tips/dones within about an hour (for example, several tips while at one place).
- **(b) True:** the median, average and maximum curves rise slowly. Reading the CDF, about half the users have an average gap of 450 hours or more, and about 80% have a maximum gap above 167 hours (about one week). Typical users are not very active, with long pauses between posts.

Figure 5: distance between consecutive activities (km).

- **(c) True:** the curves start at about 0.36 at x = 0, meaning about 36% of users have average and maximum displacement of 0 km: all their activities are at essentially the same spot.
- **(d) True:** the maximum curve reaches only about 0.9 by 6,000 km, so about 10% of users have at least one pair of consecutive posts 6,000 km or more apart (for example, after international travel).

Privacy insight: since many users post repeatedly from the same small area, their public tips and dones can reveal where they live or work, even though their check-ins are private (Question 7).

Answer: All four statements (a), (b), (c) and (d) are correct.

Answer Key

Q.No.	Correct answer
1	Foursquare, Yelp, Gowalla, Uber (all four)
2	Pearson: linear; Spearman: monotonic
3	Spearman's rank correlation coefficient
4	Venue location, User home city
5	Very skewed; a few cities have as many as 100
6	Scatter: two numerical variables; bar: categorical data
7	Mayorships, Tips, Dones
8	Bubble plot
9	Connect with others and share physical locations and activities
10	All four statements

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